

GUELFEN Abdelheq

E-mail: hakouguelfen@protonmail.com

Phone: +213696463735

Website: <https://hakoudev.web.app>

Repository: <https://gitlab.com/hakouguelfen>

Linkedin: <https://linkedin.com/in/abdelheq-guelfen>

Experience

GlaxoSmithKline(GSK)

Master's Thesis - Analyzing Deviations in Quality Control at GlaxoSmithKline (GSK)

Fev – Jul 2025

- Developed an AI-powered web application to automate root cause identification for out-of-specification (OOS) results in pharmaceutical quality control, reducing investigation time and costs.
- Built full-stack solution using Next.js frontend, Firebase for authentication and database, and Flask backend server for API management.
- Implemented a neural network using TensorFlow to solve multiclass root cause classification, achieving 98% accuracy in predictions, with scikit-learn and pandas for data preprocessing.
- Created centralized platform enabling investigation managers, laboratory staff, and production teams to efficiently exchange data and automatically distribute corrective actions (CAPAs).
- Gained valuable insights into large-scale pharmaceutical operations and enterprise-level quality control processes.
- Delivered a solution with potential to significantly reduce GSK's expensive and time-intensive investigation procedures through automated root cause identification.

Hanuut Express [<https://hanuut.com>]

Co-Founder and Lead Developer

2022 – 2024

- Co-founded a startup and led technical development of an e-commerce platform enabling local shops to establish online presence and sell products digitally.
- Developed a cross-platform mobile application using Flutter for customer and shop management.
- Built backend API with NestJS and MongoDB, implementing JWT authentication/authorization and database operations for the platform.
- Built a real-time delivery system using Flutter with Google Maps integration, WebSocket connections for live route updates, and Firebase push notifications to alert nearby drivers of new orders.
- Gained valuable experience in team dynamics and meeting project deadlines.

Projects

Anzar Authentication Service [https://anzar_software.gitlab.io/docs/]

Open Source Project

2024 – Present

- Architected a production-ready authentication microservice in Rust, implementing dual JWT/Session strategies and MongoDB/Postgres support with (RBAC) authorization system.
- Engineered a comprehensive security layer preventing CSRF, XSS, SQL injection, and DoS attacks through rate limiting, constant-time execution, and device cookies.
- Developed SDKs for Python/Ts, containerized the service as Docker image to simplify its usage, and created documentation to facilitate integration into existing applications.
- Built a cross-platform CLI (distributed via npm, cargo, and shell installers) to scaffold configuration, generate and run database migrations, rotate signing keys, and validate service health.

Education

Master's Degree in Bio-informatics

2023 – 2025

University of Science and Technology Houari Boumediene (USTHB)

Bachelor's Degree (Licence) in Computer Systems (SI)

2020 – 2023

University of Batna 2
